



Wresting with Implementation: a Step-By-Step Guide to Implementing Entrustable Professional Activities (EPAs) in Psychiatry Residency Programs

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Competency-based education has become the dominant international framework for medical education [1]. It has not come easily. This important shift from structure and process measures (e.g., time-based rotations) to educational outcomes marks significant progress in the accountability of the medical education community to learners, faculty, and the public, yet the shift has also introduced complexity for educators [2]. For example, while the Accreditation Council for Graduate Medical Education (ACGME) milestones for psychiatry residency programs were intended to make competencies more concrete and amenable to assessment, with 22 sub-competencies, 66 threads, and over 250 milestones, implementation has been challenging for faculty and programs [3].

In response to these challenges, the concept of entrustable professional activities (EPAs) was introduced to operationalize competency-based education and, in particular, workplace-based assessment in a holistic way [4]. EPAs are units of professional practice (e.g., conducting a psychiatric diagnostic evaluation) that are executable within a given time frame, observable, and measurable [5]. EPAs require the

simultaneous integration of multiple competencies (knowledge, skills, and attitudes) in one highly constrained activity [6]. For purposes of assessment, this activity is the unit of observation. Given their alignment with the natural workflow in the clinical learning environment, EPAs provide an opportunity to facilitate competency-based implementation.

Taken together, all the EPAs for a given specialty define its core identity. EPAs have been developed for many specialties. Recognizing the potential for end-of-training EPAs in the USA, the Executive Council of the American Association of Directors of Psychiatric Residency Training (AADPRT) created the EPAs for Psychiatry Task Force in 2014. The Task Force employed a rigorous, multi-stage process to develop EPAs that were essential, clear, and representative. This process, the most comprehensive to date for psychiatry, yielded 13 end-of-training EPAs [7].

With this list of EPAs for psychiatry, residency programs have the opportunity to implement EPAs as an assessment strategy. In 2020, the AADPRT Assessment Committee developed an EPA toolkit to assist programs with implementation [8]. This toolkit is meant to serve as a general guide for psychiatry residency educators. In the AADPRT EPA toolkit, the committee proposes 11 steps to EPA implementation which we summarize here.

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Step 1: Make the Case for EPAs

Many residency programs struggle with how to make milestones-based assessments in the workplace practical for front-line faculty and how to align assessments with both the curriculum and the workflow of busy, clinical learning environments. Specifically, clinical competence may not be represented by the sum of all milestones and competencies may not exist as attributes independent of their clinical context and content. Some have argued that the units (i.e., milestones or

sub-competencies) of assessment are too numerous, too granular, and/or too abstract for educators to meaningfully evaluate [3]. EPAs offer a useful approach to workplace-based assessment in competency-based education [5, 9]. Trust is a central concept for safe and effective healthcare and is an important component to teaching and learning. The ultimate goal in GME is for residents and fellows to be entrusted to do professional activities independently. What makes EPAs so intuitive for front-line educators and programs is that we are making entrustment decisions about trainees implicitly all the time [1]. EPAs bring a more explicit framework to this implicit process. EPAs also focus on the units of work that psychiatrist do in the workplace and thereby make competencies and milestones meaningful for health practice [2].

Step 2: Identify EPAs as an Assessment Framework

In developing a program of workplace-based assessment, psychiatry programs must first choose the framework for assessment. The two most common developmental frameworks for operationalizing competency-based medical education are milestones and EPAs. Each has the potential to guide the development of workplace curricula [5], and the two frameworks are complementary. EPAs combined with sub-competencies and milestones can create a meaningful framework for workplace-based assessment in healthcare. EPAs can be incorporated into a program's preexisting master map of assessment strategies and tools, also known as programmatic assessment [10].

Step 3: Adopt and Adapt List of Psychiatry EPAs

In 2018, the AADPRT EPA Task Force developed a list of essential EPAs for psychiatry (Table 1). In developing this list, the task force used a three-stage process, including a national Delphi survey of experts in education and assessment [7]. The 13 proposed EPAs were meant to be a starting point for national experimentation. The EPA Task Force encouraged programs to experiment with and adapt the EPAs in order to align with local contexts and missions. For example, programs may choose to lump or split EPAs in various ways:

- Disease (e.g., manage bipolar illness)
- Setting (e.g., diagnostic psychiatric interview in the emergency department)
- Treatment modality (e.g., initiate treatment with clozapine)
- Patient acuity/complexity (e.g., patients with one defined problem versus patients with multiple problems)

The AADPRT EPA Task Force also recommended differentiating between EPAs that should be required nationally versus those that might be required only by a specific institution (e.g., telepsychiatry, conducting ECT). Programs may add additional EPAs that are required, elective (e.g., interpersonal psychotherapy, transcranial magnetic stimulation), and/or aspirational.

Step 4: Choose an Entrustment Scale

A variety of entrustment scales have been proposed for EPAs. The original ten Cate entrustment scale consisted of five levels of entrustment [4]. Since the original ten Cate entrustment scale, other scales have been proposed. Some scales have been designed specifically for medical students [11]. Other scales focus more on procedures, asking raters to reflect retrospectively on how much the learner participated in the task [12].

For learners at the graduate medical education (GME) level, we propose the following scale:

- Level 1: Co-treat—attending demonstrates and/or co-treats
- Level 2: Direct full—attending observes and provides proactive supervision
- Level 3: Direct partial—attending only in room for a portion of the visit to verify key findings
- Level 4: Indirect—attending nearby and available as/if needed
- Level 5: Independent—ready for unsupervised practice

Step 5: Make the Use and Purpose Clear and Transparent

Entrustment decisions (and associated feedback) of a professional activity are often categorized in two ways: *ad hoc* entrustment decisions and *summative* entrustment decisions [13]. *Ad hoc* entrustment decisions happen every day, are usually made by individual supervisors, and pertain to immediate permission for the trainee to act in a specific, often limited context. *Summative* entrustment decisions are grounded in more systematic observation and lead to lasting permission to act under a specified level of supervision. They are comparable to the driver's license that formalizes permission to drive unsupervised from that point onwards.

Entrustment decisions can be determined through a variety of assessment methods (e.g., direct observation, documentation review, multisource feedback, in-training examination, etc.) to form different kinds of judgment (e.g., formative growth plans or summative advancement decisions). There should be transparency and clarity about how the data

Table 1 List of psychiatry EPA titles and examples

EPA title	Examples of the activity
Manage psychiatric patients longitudinally	• A resident is able to manage the longitudinal treatment of patients in a variety of setting such as acute hospitalization or ambulatory care. • A resident is able to manage the individual patient tasks as well as triage the multiple and sometimes competing tasks of a patient panel.
Manage psychiatric emergencies	• A resident is able to assess and manage psychiatric emergencies, including suicidality and/or homicidality. • A resident is able to perform a suicide and violence risk assessment and develop an appropriate management plan.
Conduct psychiatric diagnostic evaluation	• A resident is able to perform a diagnostic evaluation, including the elements of data gathering, mental and cognitive exam, assessment and formulation, and treatment planning. • A resident is able to conduct a diagnostic evaluation across all major psychiatric diagnostic categories, the spectrum of presentations, adult patients across the lifespan, and in a variety of settings (e.g., inpatient, outpatient, emergency).
Manage a patient's psychiatric conditions with medications	• A resident is able to perform the central task of medication management, including the essential tasks of a medication visit or follow-up whether office-, tele-, or video-based. • A resident is able to take a focused interval history and review of symptoms, use measurement-based care, develop an assessment and treatment plan, and provide referrals as needed. • A resident is able to perform medication selection and titration as well as maintaining a therapeutic alliance, adverse effects, substance misuse and adherence, and engaging the patient in all aspects of care.
Manage involuntary commitment and treatment	• A resident is able to manage clinical situations in which involuntary (civil) commitment and involuntary treatment are required for patient care. • A resident is familiar with local statutes which allow the state, in certain clinical situations, to deny patients the right to refuse treatment. • A resident applies legal standards appropriately and interface with legal authorities in petitions, proceedings, and testimony.
Assess and manage decision-making capacity	• A resident is able to assess and manage a patient's capacity to make decisions. • A resident is able to describe local statutes around decision-making capacity, apply those standards to clinical situations, and participate in petitions, proceedings, and testimony when appropriate.
Manage transitions in care	• A resident is able to provide safe and effective handoffs and transitions in care as both the sender and receiver. • A resident is aware that triggers for these transitions include a change in level of care, end of shift or rotation, patient initiated change in provider, and the provider ending their relationship with the patient's insurer or clinic.
Provide psychiatric consultation to other clinicians or services	• A resident is able to provide a psychiatric consultation across the spectrum of clinical settings and specialties. • A resident is able to assist another healthcare clinician(s) in providing safe and effective care to patients with psychiatric symptoms and illnesses via accurate, timely, and helpful guidance.
Provide supportive psychotherapy	• A resident is able to use techniques of supportive psychotherapy when treating patients with a wide range of presenting complaints and diagnoses.
Provide cognitive behavioral therapy	• A resident is able to use basic elements of cognitive behavioral therapy when treating patients with common presenting disorders (e.g., depression and/or anxiety disorders).
Provide psychodynamic psychotherapy	• A resident is able to demonstrate a basic foundation in psychodynamic theory and technique and should be able to integrate these principles and techniques into the psychotherapy treatment of patients when indicated.
Lead an interprofessional healthcare team	• A resident is able to lead an interprofessional team. • A resident is able to maintain focus on the principal goals or purpose, align with the broader mission, understand the roles of others, appreciate group dynamics, deploy medical expertise, promote shared decision-making, respect diversity, advocate for the patient, and serve as an agent of change.
Apply quality improvement methodologies to one's patient panel or clinical service	• A resident is able to apply basic quality improvement techniques in order to improve the care that their patients receive. • A resident is able to recognize that quality improvement approaches are a means to systematically assess and improve patient care and clinical outcomes.

A full description of each EPA is available upon request

generated by EPAs will be used. In ad hoc entrustment decisions, front-line faculty may use the decision for formative assessment. In coaching sessions, the data, especially when aggregated, may be used to create improvement plans. In summative entrustment decisions, a Clinical Competency Committee (CCC) may determine the level of entrustment for a particular activity based on a synthesis of multiple data points and use that level to inform advancement decisions.

Step 6: Select the Rotation(s) or Context in Which to Implement EPAs

While a program should consider the comprehensive list of essential EPAs, educators may select only certain EPAs for specific rotations or clinical learning environments. For example, on an inpatient psychiatry service, the following EPAs may be most relevant:

- Conduct a psychiatric diagnostic evaluation
- Manage the psychiatric patient longitudinally (during a hospitalization)
- Manage a patient's psychiatric medications
- Manage transitions of care

It can be a useful exercise for programs to select specific EPAs for each rotation or clinical learning environment. While not all EPAs are relevant to a given rotation, a given EPA will be relevant to multiple rotations, contexts, and environments.

Step 7: Construct your EPA Assessment Strategy

The most common workplace-based assessment tools include multi-source feedback (patients, staff, peers), chart stimulated recall, direct observation, end-rotation global feedback, portfolios, and practice-based audits. Assessment tools should collect both quantitative and narrative data. When aggregated over multiple observations, quantitative data can provide a powerful signal of the direction a trainee is going with respect to competency acquisition. Narrative comments add explanation and guidance about what was done well or not so well that is essential for the learner and CCC to have in order to place the scores in context [14].

Programs must design their assessments so that each primary EPA is assessed multiple times by at least two different methods. In addition, programs should emphasize a given assessment method in proportion to the importance of the competencies or EPA that it assesses. The assessment strategy should indicate to the learner what the program values. Too often, programs assess what is easy to assess rather than what

is important to assess. This can lead stakeholders to experience workplace-based assessment as trivial and intrusive.

Moreover, the EPA assessment strategy should utilize multiple types of tools, each administered multiple times with multiple assessors [10]. The use of “multiples” helps overcome important sources of bias. Multiple methods help to compensate for the limitations of any one technique. Multiple assessments of a given competency or EPA help overcome “one offs” (e.g., resident performing atypically) and to develop a more stable and complete view of a trainee's skill. Multiple assessors compensate for the biases (e.g., halo effects, leniency) that influence any given faculty member's judgment.

Several psychiatry-specific tools have been published. For example, the Psychopharmacotherapy-Structured Clinical Observation (P-SCO) is the direct observation and feedback tool in psychiatry with the most evidence for validity. The P-SCO was developed to assess the EPA of a follow-up (medication management) visit [15]. The P-SCO includes a checklist of the essential tasks of a medication visit plus space for narrative comments. The feasibility and utility of implementing the P-SCO has been demonstrated in several studies [15, 16]. Additional research has reported evidence for validity of the P-SCO with respect to its content, internal structure, and association with other variables [17, 18]. Moreover, the P-SCO has been found to generate high-quality narrative comments that are behaviorally specific and balanced between being corrective and re-enforcing. The comments either add unique content or elaborate on the “why” behind a low or high quantitative score [14, 15].

Evidence for validity has also been developed for the EPA of a psychiatric diagnostic interview [19]. Recently, a mobile app that assesses EPAs in psychiatry has shown evidence of feasibility and validity [20]. Completion of a single assessment with the EPA app took on average 72 s and generated one high quality, specific corrective comment. Several residency programs have developed mobile apps to assist in entrustment decisions on specific EPAs in the workplace. There are many more examples in the AADPRT EPA toolkit, including EPA-based global assessment forms for use at the end of a learning period.

Step 8: Remember that Implementation of EPAs Is an Iterative Process

Implementation of any assessment program in the workplace will encounter significant challenges. It is to be expected. Supervisors and trainees often are inadequately trained, do not understand the purpose of assessments, and interact in settings with insufficient time for assessment and/or feedback. Studies have identified additional barriers such as brief and frequently changing supervisory relationships that do not

permit the development of strong educational alliances, poorly designed assessment tools that are experienced as burdensome, and low-quality, unidirectional feedback that is not credible to the trainee. Faculty ratings can vary widely [21], and trainees often perceive assessment as summative even when it was intended as formative.

Effective assessment programs will bundle together multiple strategies such as protected faculty time for observation and feedback, longitudinal supervisory relationships, bidirectional feedback models, and ongoing faculty and resident training [21]. In addition, programs must carefully choose the platforms for their assessment tools. The principal choices are paper-based versus mobile apps. A series of studies using implementation science to identify the barriers and facilitators of engagement with both the EPA app and a paper-based assessment yielded several insights [16, 21, 22]. First, it is critical that the assessment tool be developed utilizing an iterative, user-centered process in order to minimize cognitive load and ensure ease of use (feasibility) and high-quality data. Well-designed, easy-to-use tools are much more likely to be used in the manner intended. Second, careful attention should be given to how the assessment tools, whether electronic or paper, are embedded within clinical workflows to ensure feasibility [23]. Alpha and beta testing can help identify features of the tool and/or workflow that either facilitate or impede engagement. Third, these studies found that faculty and residents value the quickness of the mobile app and how the app encouraged the faculty member to distill feedback into a succinct point. On the other hand, both faculty and residents appreciated how the paper-based form with a checklist generated more thorough, systematic feedback. Faculty, however, indicated that they would resent completing the checklist on a smartphone. These findings suggest that there is a role for both types of platforms for workplace-based assessments.

Here are some example questions to consider in a successful EPA implementation:

- Which EPA(s) to select in your curriculum and why?
- What assessment methods (e.g., direct observation, global assessment) to use to come to ad hoc or summative entrustment decisions?
- What are the necessary steps to incorporate EPAs into a workflow in the clinical learning environment?
- How is information from this assessment recorded?
- How is information from this assessment used?
- What feedback is provided to learners from the assessment?
- How to orient, on an ongoing basis, learners to the assessment?
- How to orient, on an ongoing basis, faculty to the assessment?
- What is the evidence that this was an effective assessment?

Step 9: Evaluate the Impact of Your Assessment Program on the Learners, Faculty, and Program Culture

The implementation of any assessment program, including EPAs, will influence the culture of a program and all stakeholders, including the learners, faculty, and program. Do learners value the feedback that was generated by the assessment program? Does the assessment program promote a growth or performance mindset? Is the data usable by the CCC? What and how a program chooses to assess is a powerful statement about priorities. Programs want to make sure that their assessment program is aligned with their values and vision.

The current international consensus on evaluating an assessment program has identified the following framework and related questions a program should reflect on following an EPA implementation phase [10, 24]. Many of these questions are common to all new assessment rollouts.

- **Validity**—Is the EPA measuring what you want it to measure? Do faculty have similar conceptions of what competence looks like for each EPA? Do faculty understand the entrustment scale?
- **Reliability**—Is the assessment producing a consistent result across users?
- **Acceptability**—Do learners and faculty accept and adopt the new method? Or do they resist it?
- **Educational impact**—Do EPAs have the desired educational outcome in the clinical learning environment?
- **Cost**—How resource-intensive is the new tool to implement?

Step 10: Promote Faculty Development

In our experience, a program can never spend enough time in faculty development when it comes to rolling out a new assessment framework or tool. It is essential to orient and teach faculty and learners on how to use EPAs in the clinical learning environment prior to rolling out the new approach. Front-line faculty often appreciate how simple EPAs are relative to other assessment frameworks as well as how aligned EPAs are with their everyday decision-making about trainees. Yet, recent studies have shown that faculty do not understand or interpret EPAs in a consistent fashion [22]. In faculty development, it is important to focus on the performance dimensions (i.e., what does the competent performance of the task look like in behavioral terms), frame of reference (e.g., compared to what?), the framework being used, how to complete the assessment in the course of one's everyday workflow, and how data from that assessment will be monitored and used

towards learner development. Assessment data can be both formative and summative.

We encourage programs to provide an in-service to their faculty and learners when rolling out a new assessment tool and on an ongoing basis. The evidence is, unfortunately, clear that one-time faculty and learner trainings are inadequate [25]. The trainings must be repeated and interactive with opportunities to practice and calibrate.

Step 11: Link EPAs to Milestones

Many programs are interested in how front-line assessments can ultimately link to the summative decisions made by Clinical Competency Committees (CCCs). Specifically, CCCs are interested in how EPAs can link to milestones. ACGME milestones are stages in the development of specific competencies. Milestones may link to a supervisor's entrustment decisions. The AADPRT EPA Task Force proposed a guide to how EPAs link to the ACGME Milestones in Psychiatry [7].

Programs are encouraged to refer to their specific assessment platform (e.g., MedHub, New Innovations, E*Value) for the possibility of linking EPAs to milestones on the back-end such that a given rating by a front-line faculty member using EPAs will be mapped onto milestone scores for the CCC. In our experience, most programs base their milestone ratings on inference rather than specific linkages, although linkages can be a useful starting point in a CCC discussion.

Summary

EPAs offer a practical approach to workplace-based assessment. EPAs offer substantial opportunities for learners, faculty, and CCCs in formative and summative assessments. In conjunction with the milestones, EPAs offer a complementary approach to competency-based education. The AADPRT EPA Toolkit aims to help programs implement EPAs successfully by offering a step-by-step guide.

Compliance with Ethical Standards

Conflict of Interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

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